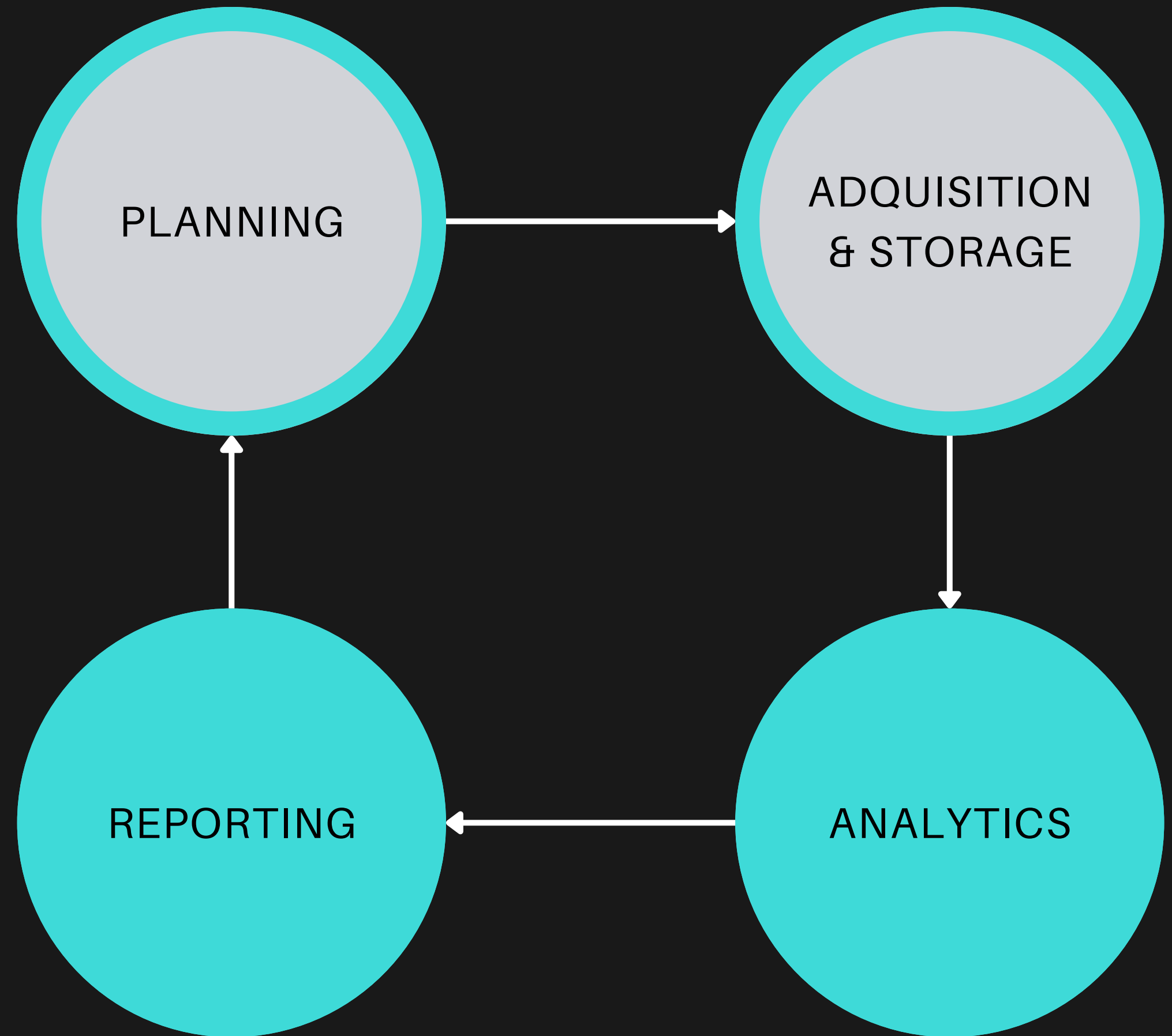


GAME ANALITICS

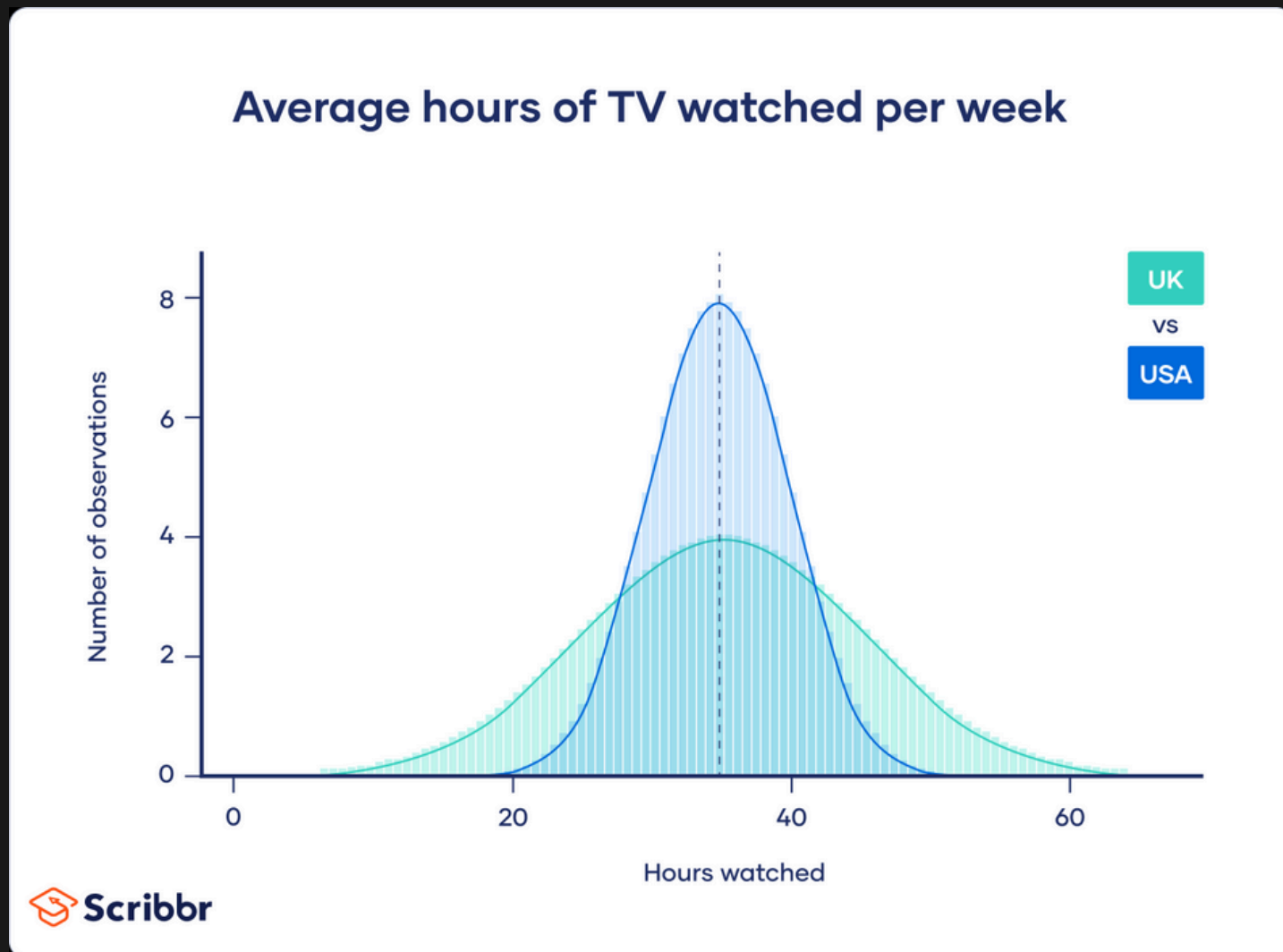
Confidence interval



CONFIDENCE INTERVAL

A confidence interval (CI) is a **range of estimates** for an unknown parameter. A confidence interval is computed at a designated **confidence level**; the 95% confidence level is most common, but other levels, such as 90% or 99%, are sometimes used

CONFIDENCE INTERVAL

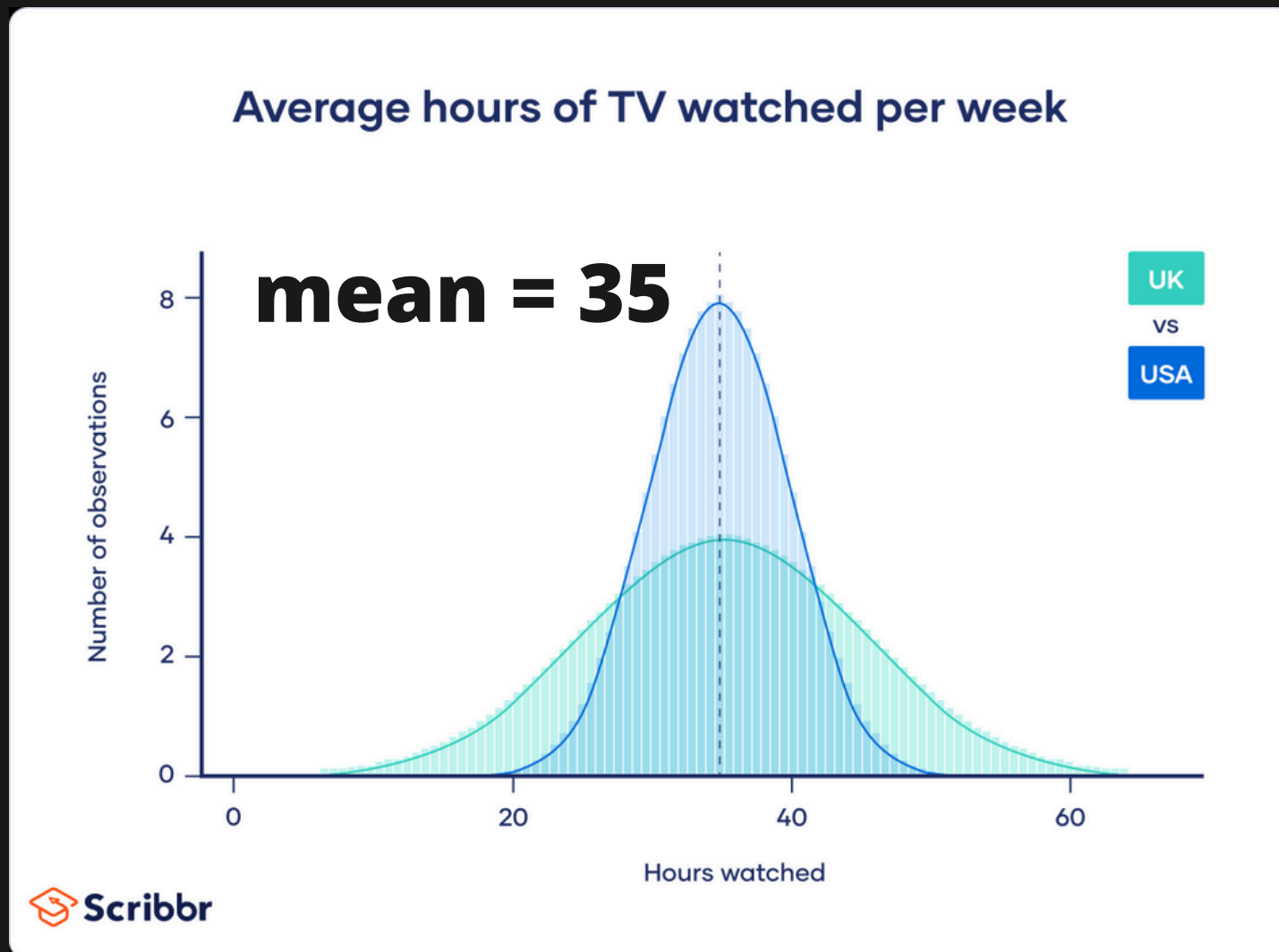


1. The point estimate you are constructing the confidence interval for
2. The critical values for the test statistic
3. The standard deviation of the sample
4. The sample size

GAME ANALITICS

Confidence interval

MEAN VALUE



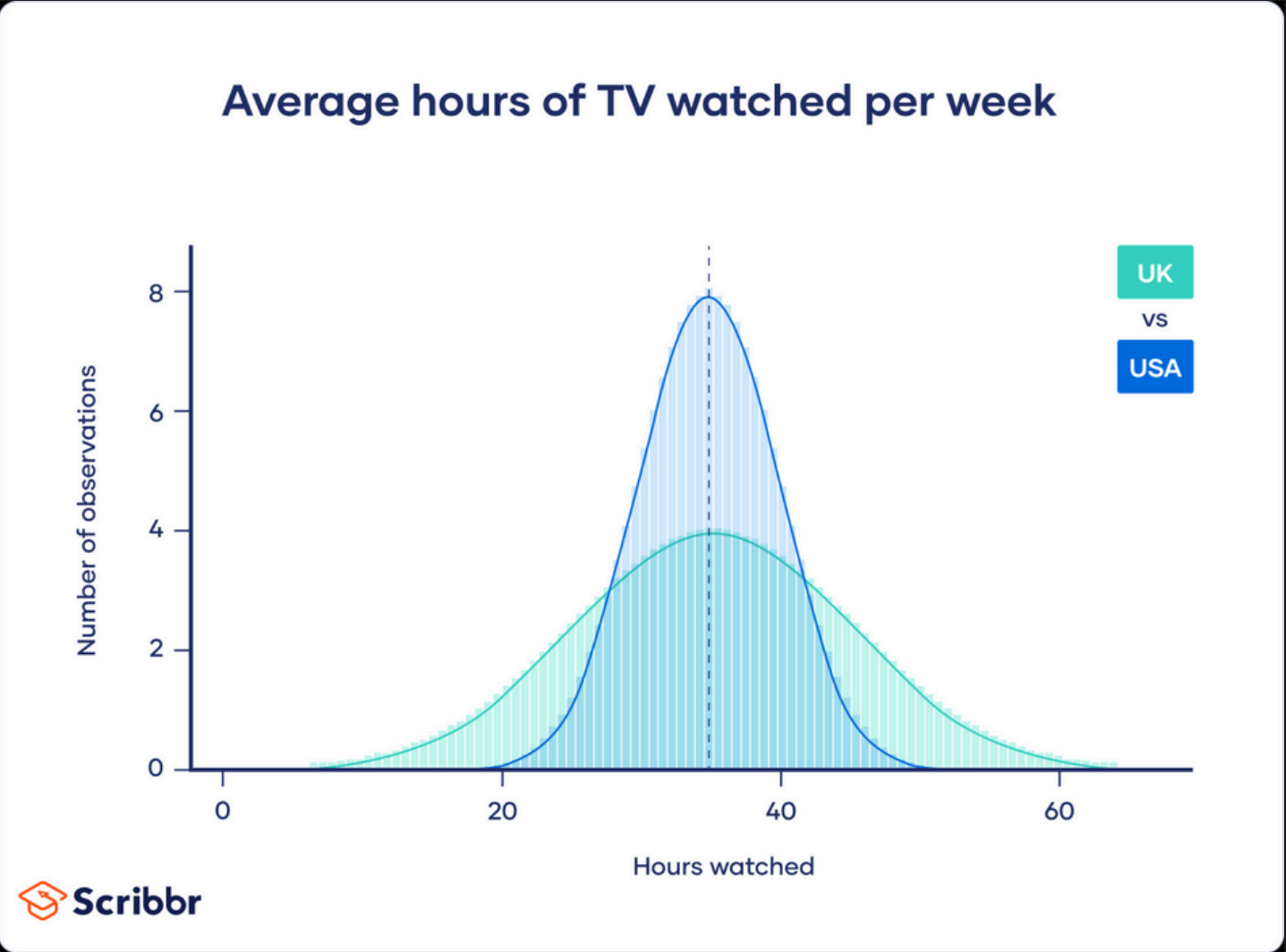
1. The point estimate you are constructing the confidence interval for
2. The critical values for the test statistic
3. The standard deviation of the sample
4. The sample size

THE POINT ESTIMATE OF YOUR CONFIDENCE INTERVAL WILL BE WHATEVER STATISTICAL ESTIMATE YOU ARE MAKING

GAME ANALITICS

Confidence interval

Z-VALUE



1. The point estimate you are constructing the confidence interval for
2. The critical values for the test statistic
3. The standard deviation of the sample
4. The sample size

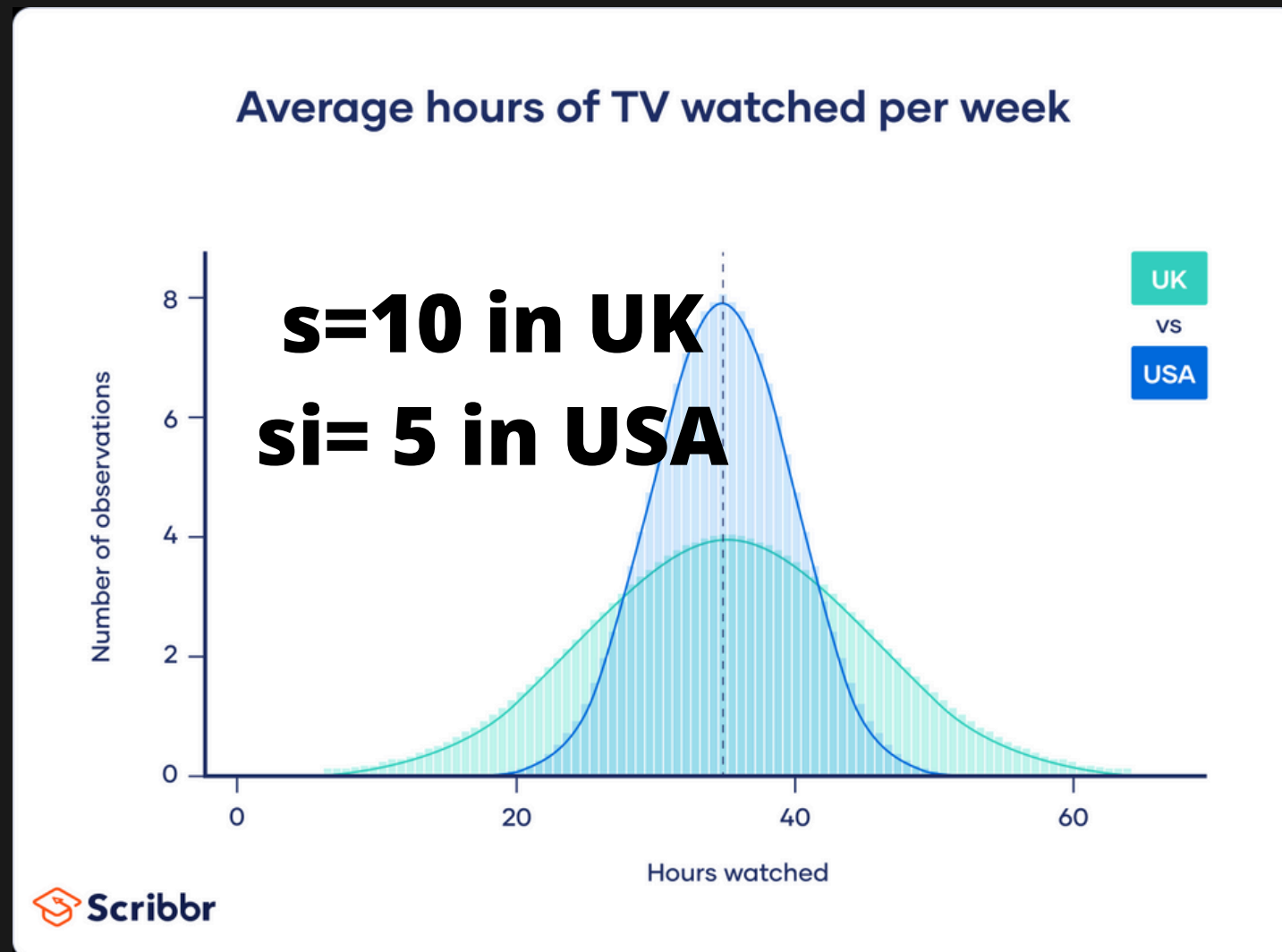
Confidence level	90%	95%	99%
alpha for one-tailed CI	0.1	0.05	0.01
alpha for two-tailed CI	0.05	0.025	0.005
z statistic	1.64	1.96	2.57

GAME ANALITICS

Confidence interval

condition: $n > 30$

STANDARD DEVIATION



1. The point estimate you are constructing the confidence interval for
2. The critical values for the test statistic
3. The standard deviation of the sample
4. The sample size

SAMPLE VARIANCE IS DEFINED AS THE SUM OF SQUARED DIFFERENCES FROM THE MEAN, ALSO KNOWN AS THE MEAN-SQUARED-ERROR (MSE):

$$s^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$

THE STANDARD DEVIATION IS EQUAL TO THE SQUARE ROOT OF THE SAMPLE VARIANCE (S²):

$$s = \sqrt{(s^2)}$$

GAME ANALITICS

Confidence interval

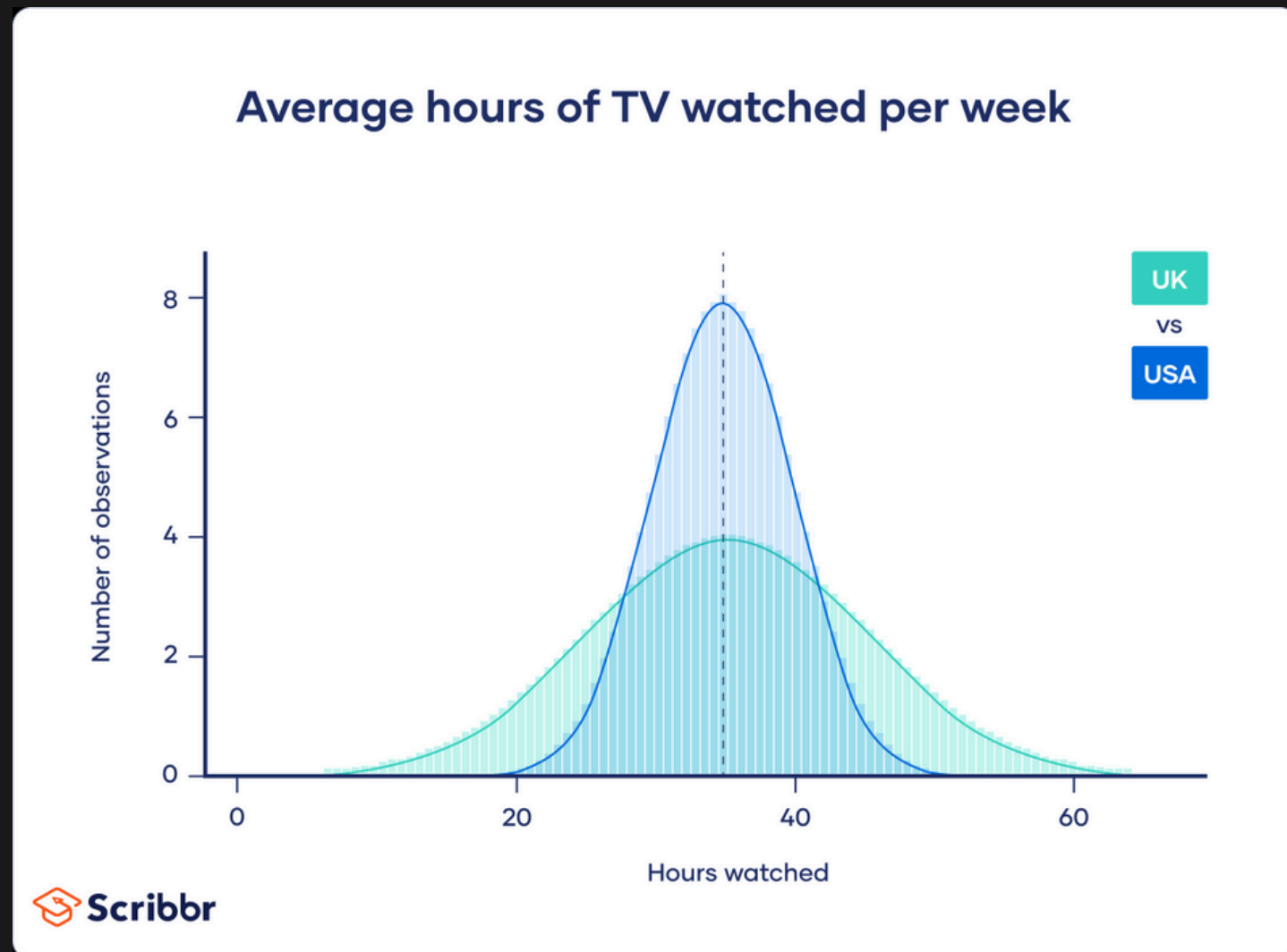
STANDARD DEVIATION

[HTTPS://WWW.MATHSISFUN.COM/DATA/STANDARD-DEVIATION-FORMULAS.HTML](https://www.mathsisfun.com/data/standard-deviation-formulas.html)

GAME ANALITICS

Confidence interval

SAMPLE SIZE



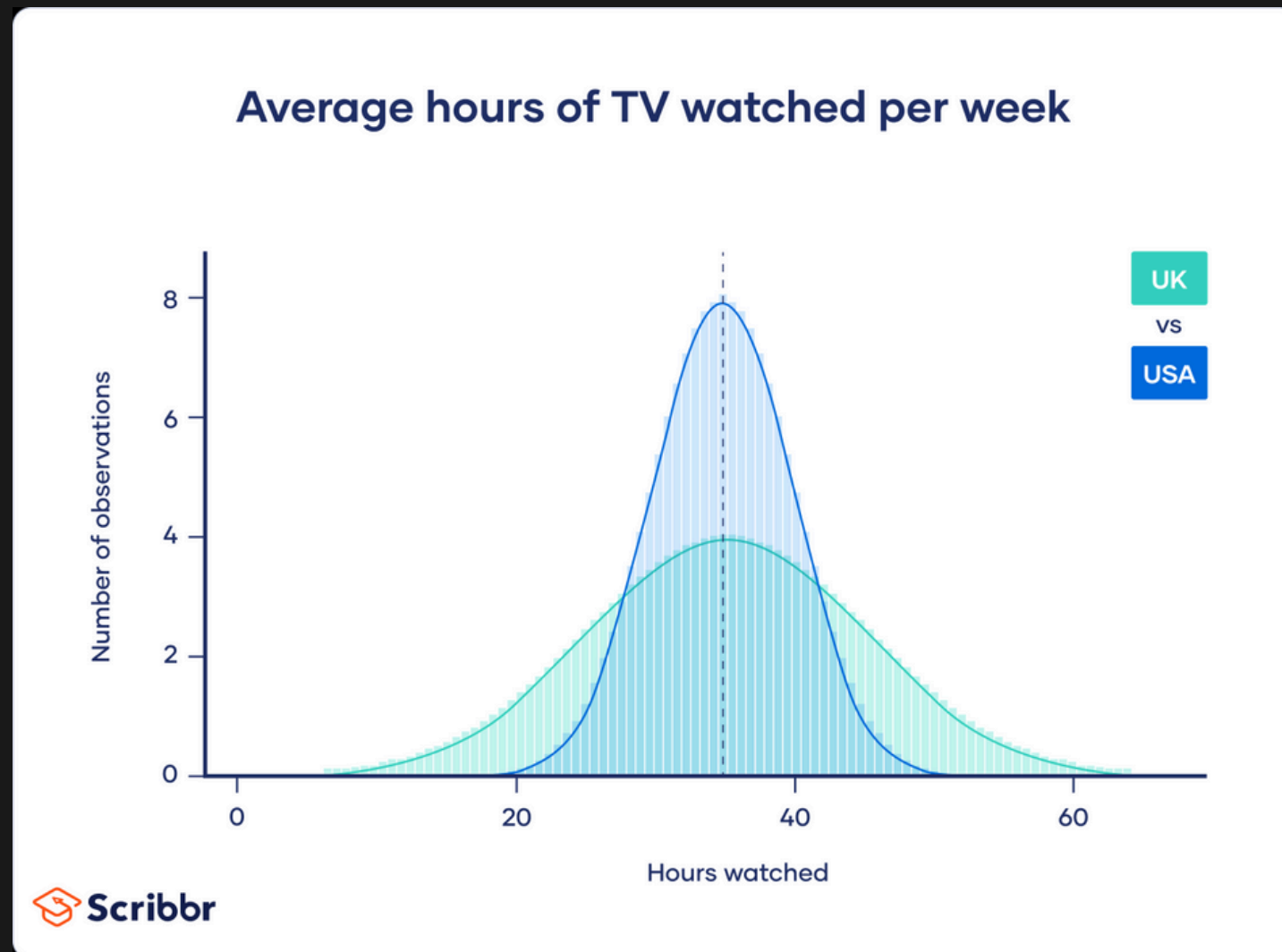
1. The point estimate you are constructing the confidence interval for
2. The critical values for the test statistic
3. The standard deviation of the sample
4. The sample size

$n=100$

GAME ANALITICS

Confidence interval

CONFIDENCE INTERVAL



THE CONFIDENCE INTERVAL FOR DATA WHICH FOLLOWS A STANDARD NORMAL DISTRIBUTION IS:

$$CI = \bar{X} \pm Z^* \frac{\sigma}{\sqrt{n}}$$

Standard error

WHERE:

CI = the confidence interval

$\bar{X}$ = the population mean

Z^* = the critical value of the z distribution

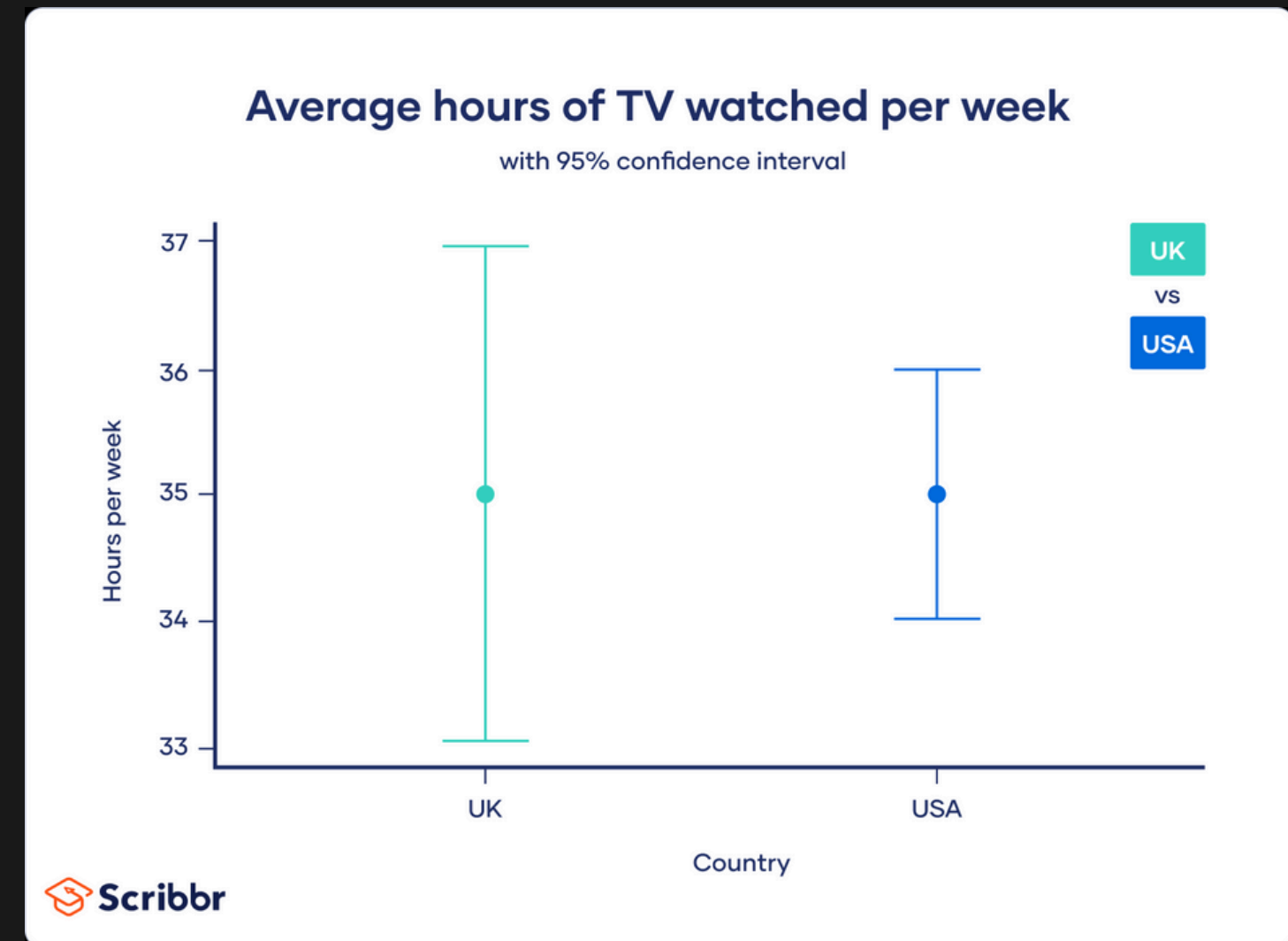
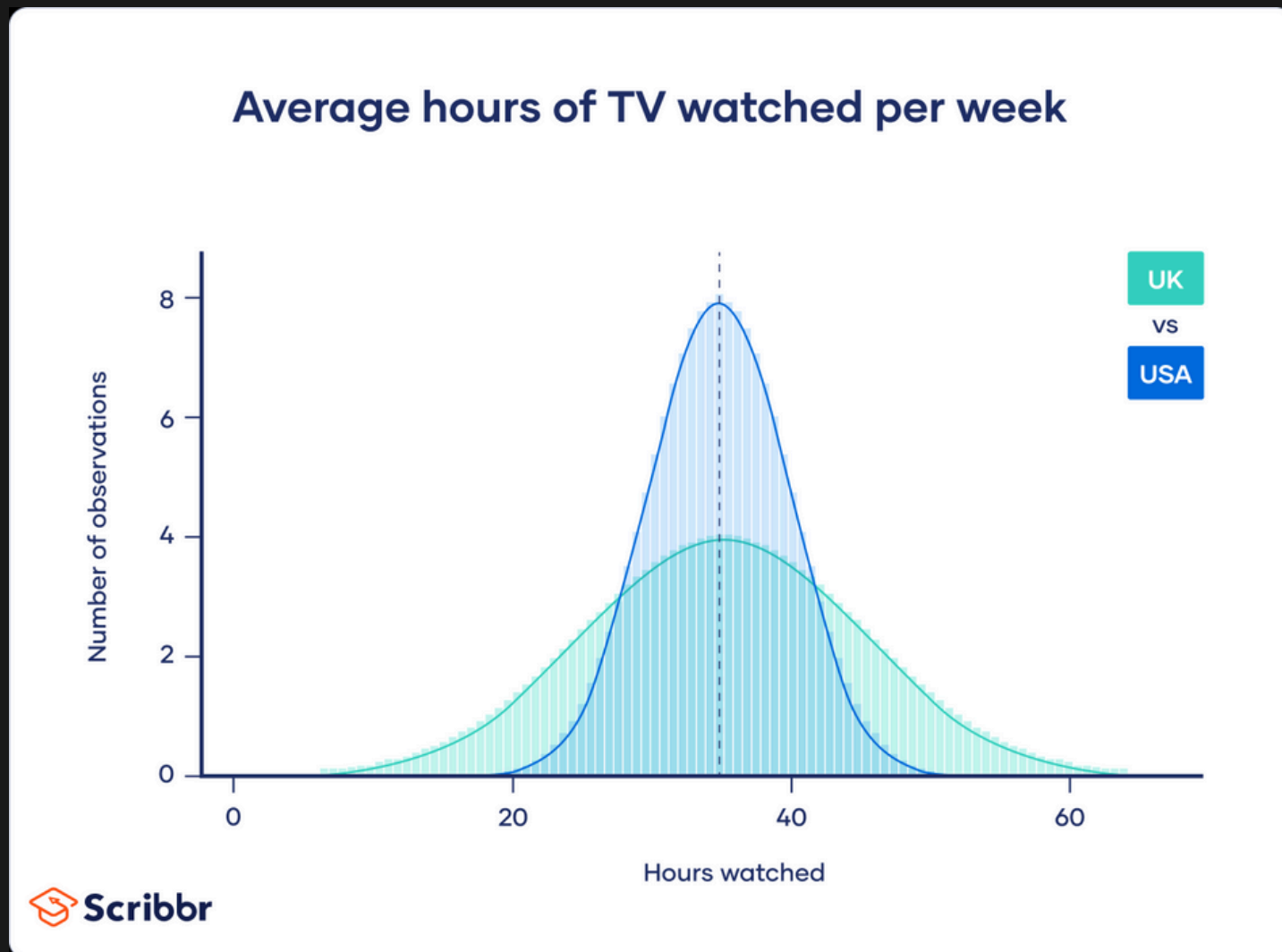
σ = the population standard deviation

$\sqrt{n}$ = the square root of the population size

GAME ANALITICS

Confidence interval

CONFIDENCE INTERVAL



GAME ANALITICS

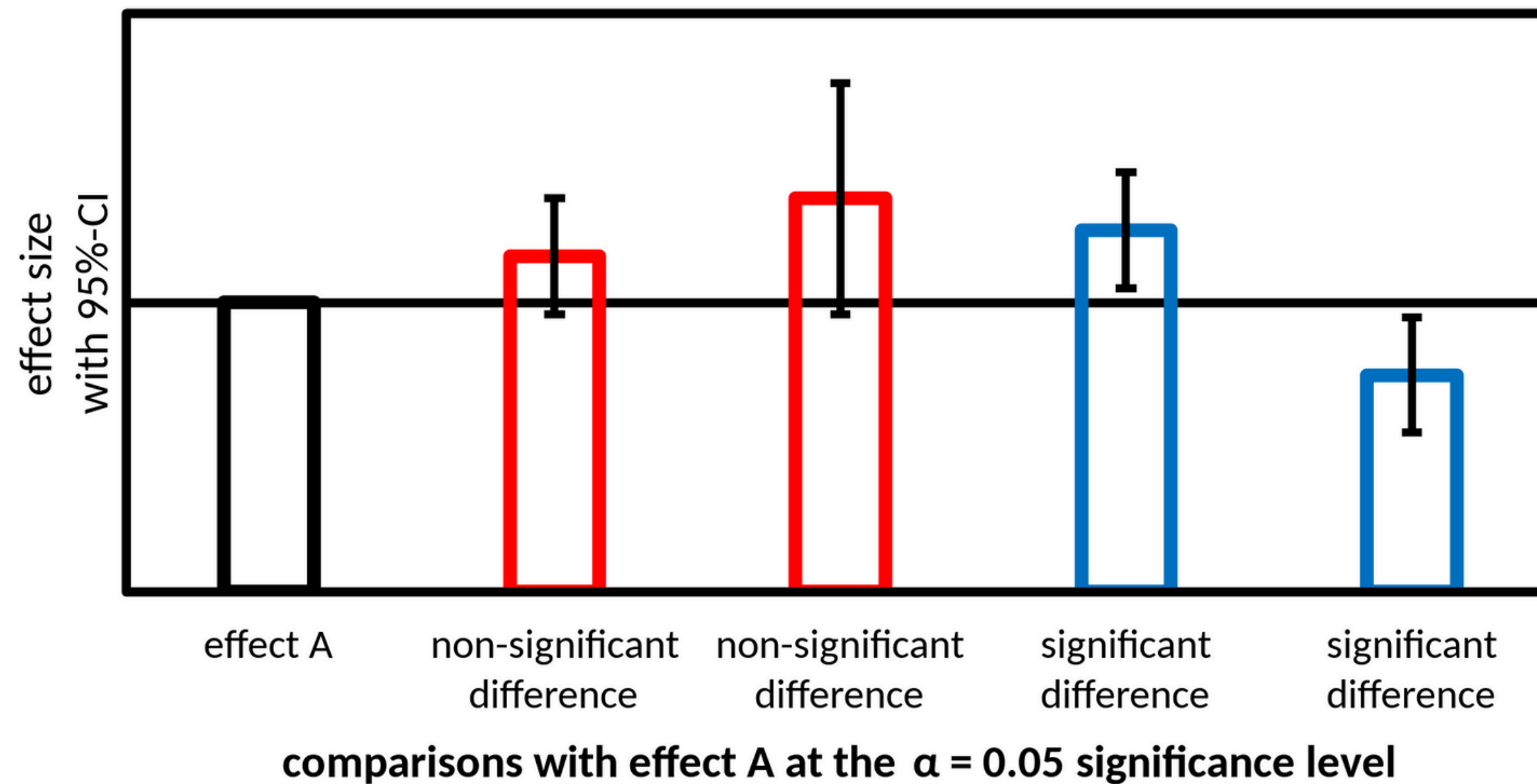
Confidence interval

Source:

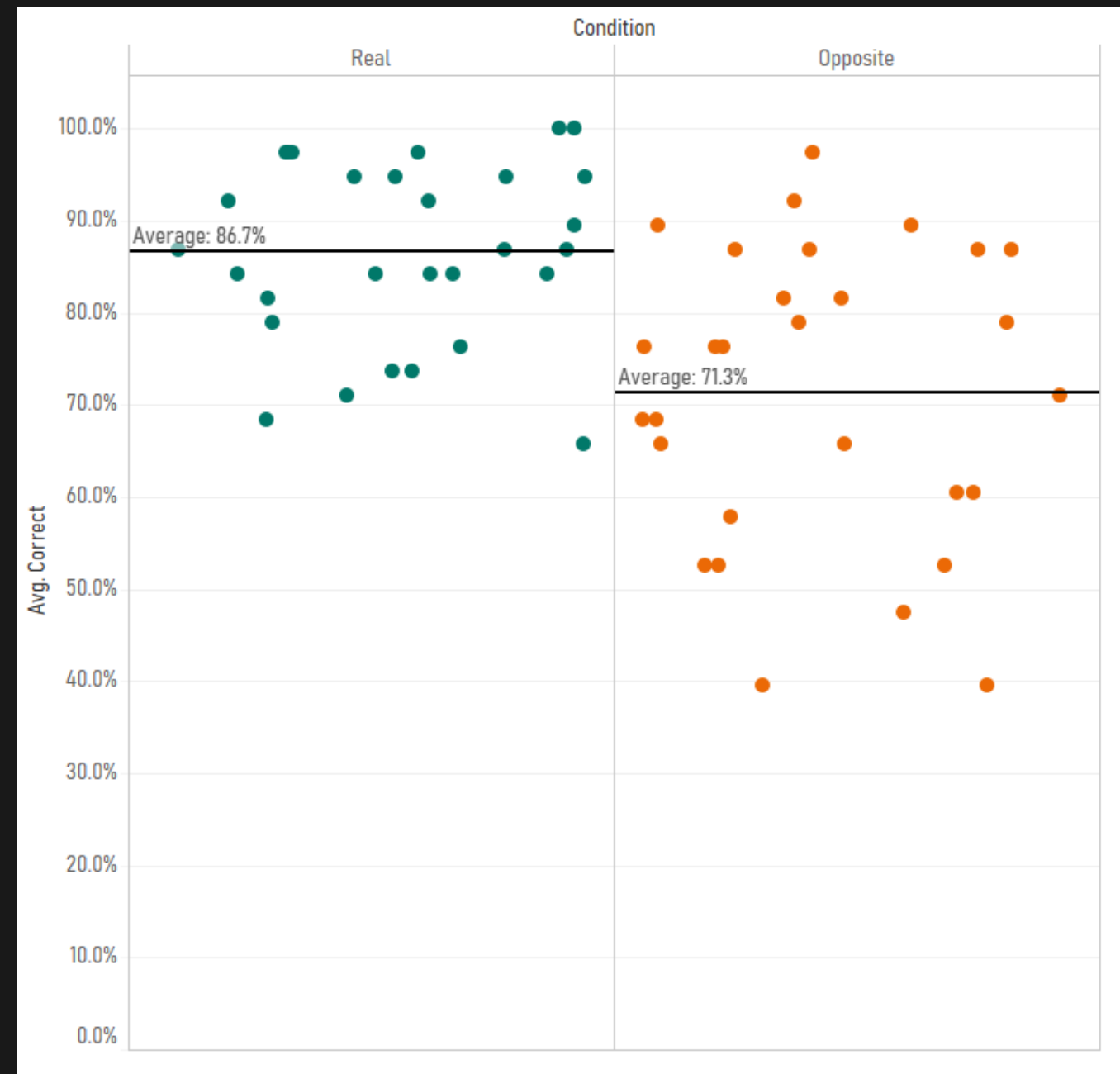
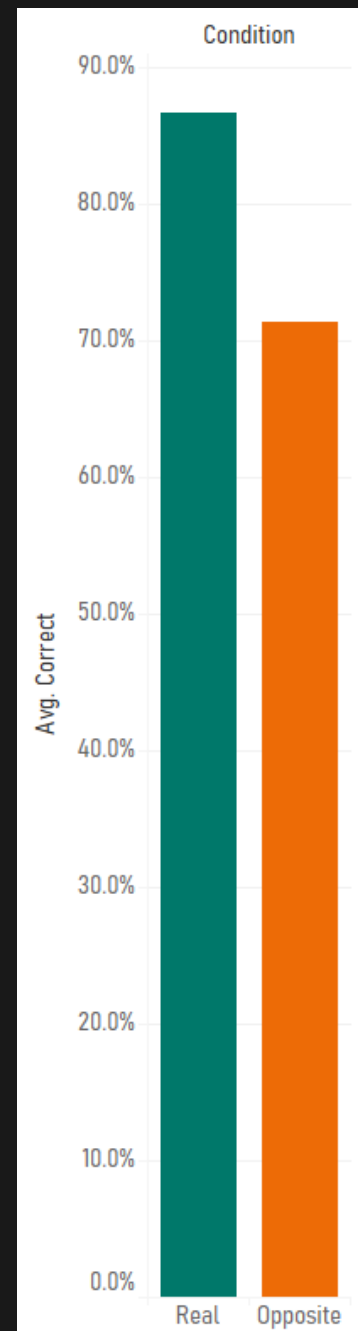


CONFIDENCE INTERVAL

the 95% confidence interval



CI - TABLEAU



GAME ANALITICS

Confidence interval

CI - TABLEAU

Dots, means, 95% CIs (data) (3)

×

```
STDEV([Correct])  
//or [Sample STDEV] or [Not sample STDEV]  
/  
SQRT(COUNTD([Participant]))
```

The calculation is valid.

8 Dependencies ▼

Apply

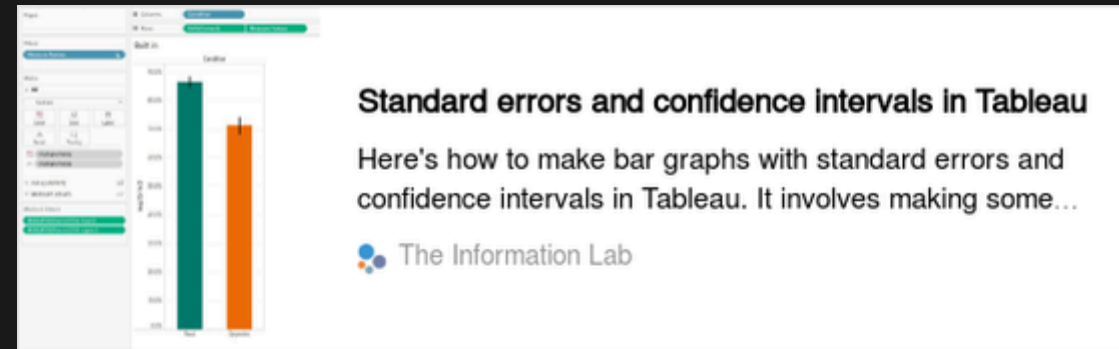
OK

**AVG([CORRECT]) - [SE]
AND
AVG([CORRECT]) + [SE]**

GAME ANALITICS

Confidence interval

CI - TABLEAU



GAME ANALITICS

Confidence interval

CI - R

```
# R program to determine the mean  
  
# Calculate the mean of the Sepal.Length  
mean_value <- mean(iris$Sepal.Length)
```

```
# Compute the size  
n <- length(iris$Sepal.Length)  
  
# Find the standard deviation  
standard_deviation <- sd(iris$Sepal.Length)  
  
# Find the standard error  
standard_error <- standard_deviation / sqrt(n)
```

CI - R

```
alpha = 0.05
degrees_of_freedom = sample.n - 1
t_score = qt(p=alpha/2, df=degrees_of_freedom, lower.tail=F)
print(t_score)
```

if you want to use z value use qnorm()

Sintaxis función qnorm

```
qnorm(p,                # Número o vector de probabilidades
      mean = 0,          # Número o vector representando la/s media/s
      sd = 1,            # Número o vector representando la/s desviación/es típica/s
      lower.tail = TRUE, # Si TRUE, las probabilidades son  $P(X \leq x)$ , o  $P(X > x)$  si F
      log.p = FALSE)     # Si TRUE, las probabilidades vienen dadas como Logaritmos
```


CI - R

```
margin_error <- t_score * standard_error
```

```
# Calculate the lower bound  
lower_bound <- mean_value - margin_error  
  
# Calculate the upper bound  
upper_bound <- mean_value + margin_error
```

CI - R - AUTOMATED

```
# R program to find the confidence interval  
  
# Calculate the mean and standard error  
model <- lm(Sepal.Length ~ 1, iris)  
  
# Find the confidence interval  
confint(model, level=0.95)|
```



How to Find Confidence Intervals in R?
Your All-in-One Learning Portal: GeeksforGeeks is a
comprehensive educational platform that empowers...
GeeksforGeeks / Jul 24

GAME ANALITICS

Confidence interval

Source:

CI - R - AUTOMATED

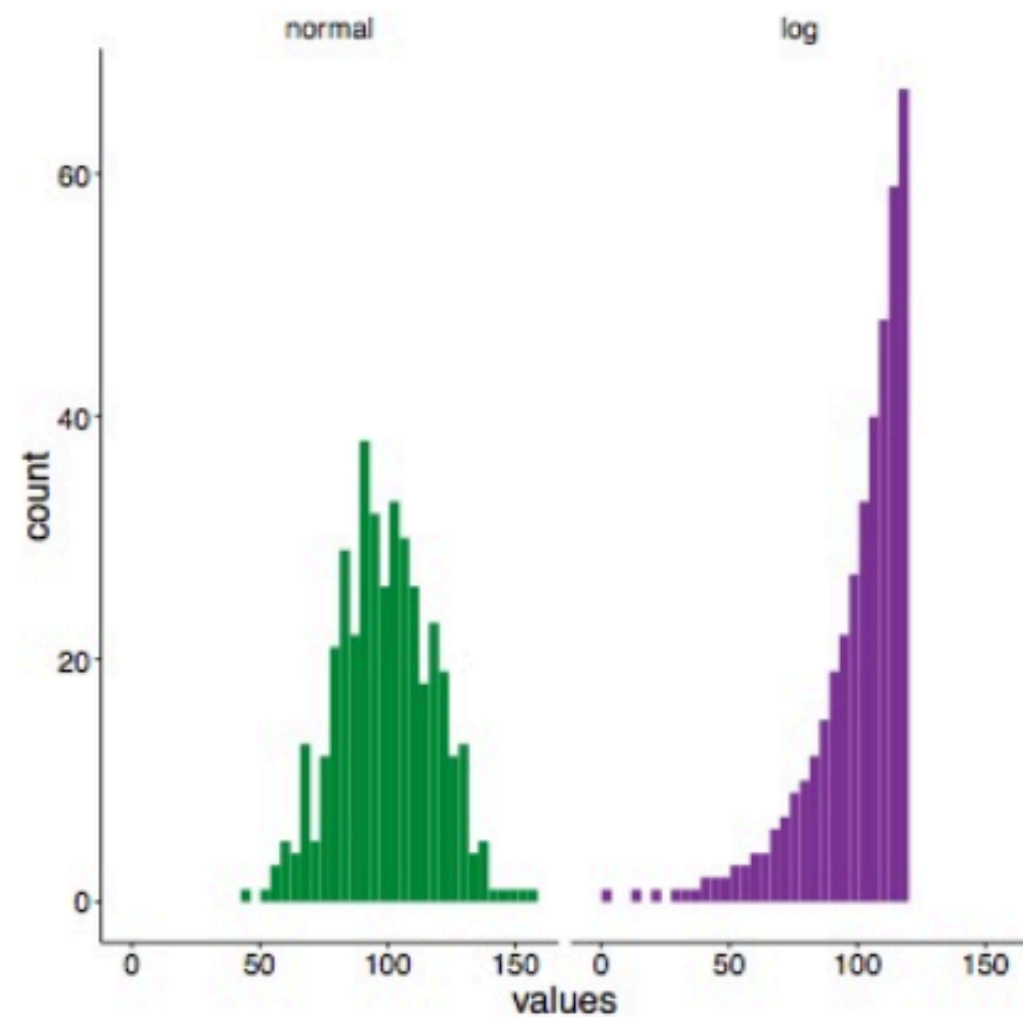


GAME ANALITICS

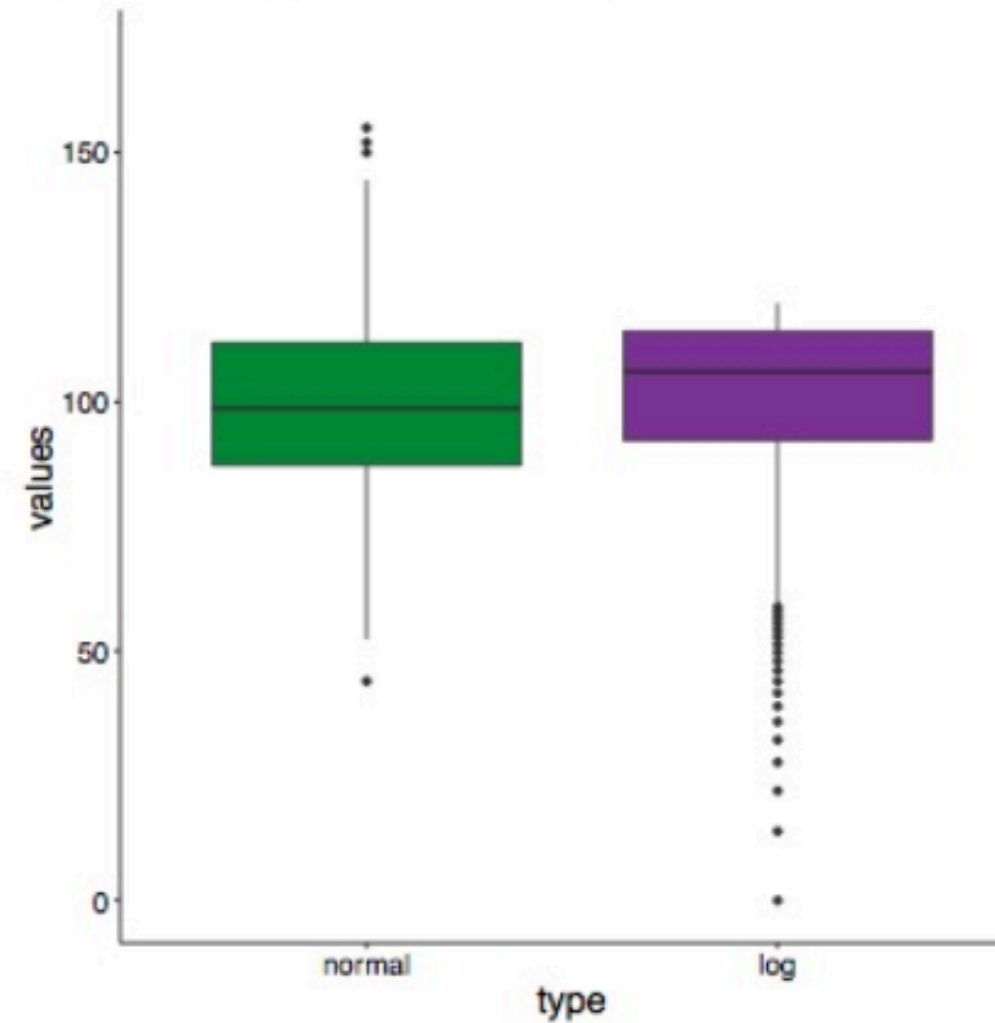
Confidence interval

CI - A WORD OF CAUTION

Clearly different...



also pretty clearly different....

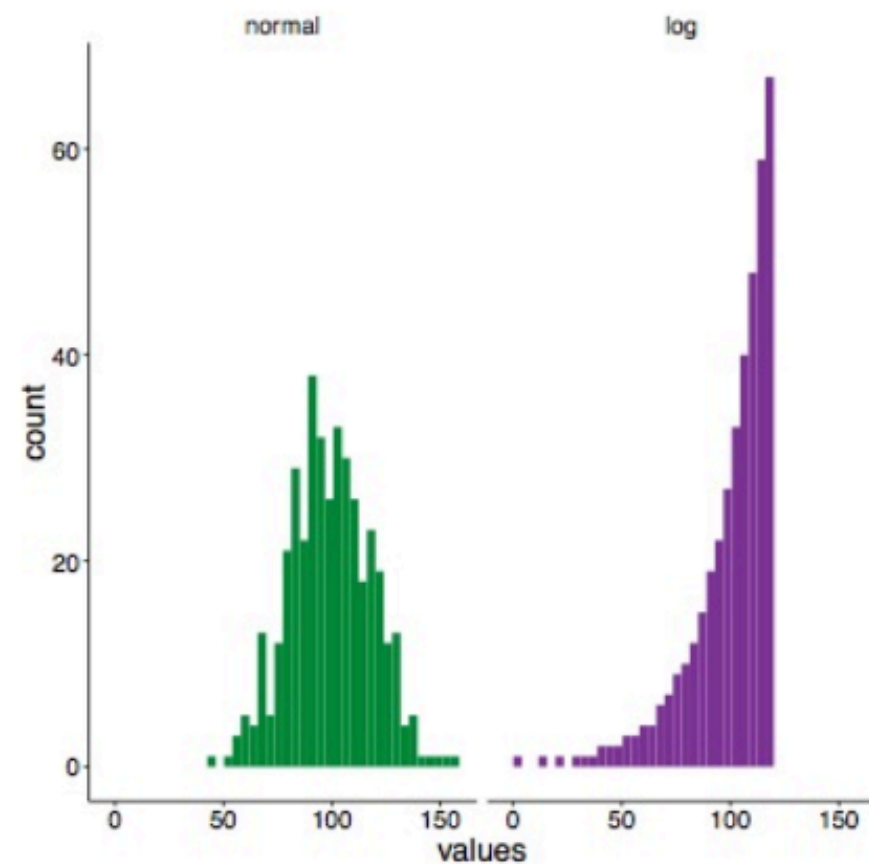


GAME ANALITICS

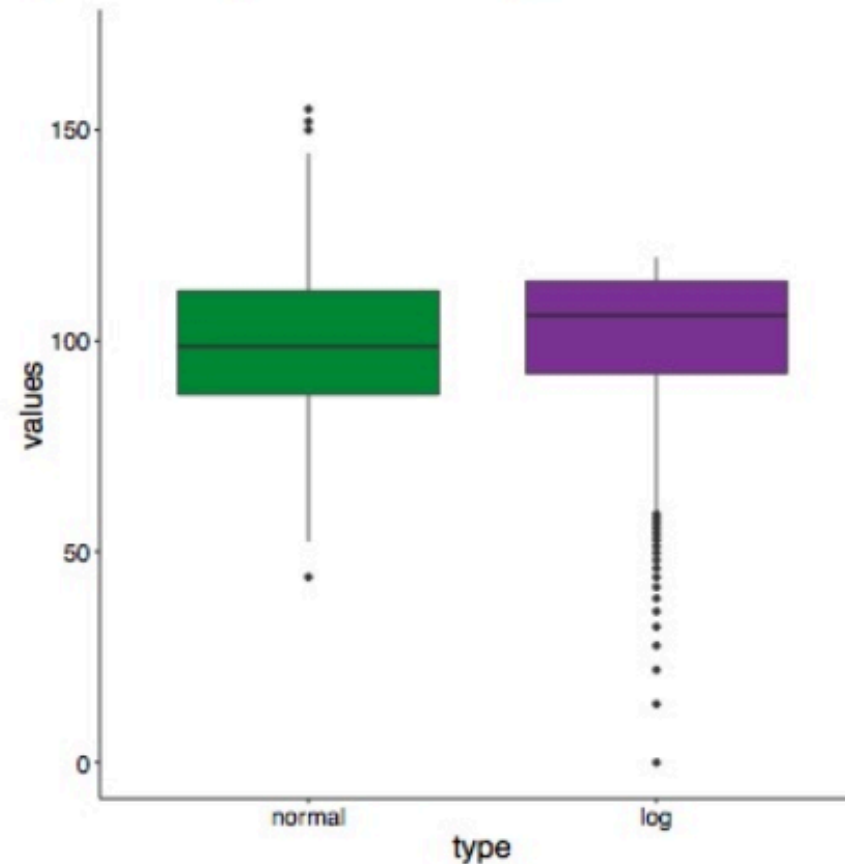
Confidence interval

CI - A WORD OF CAUTION

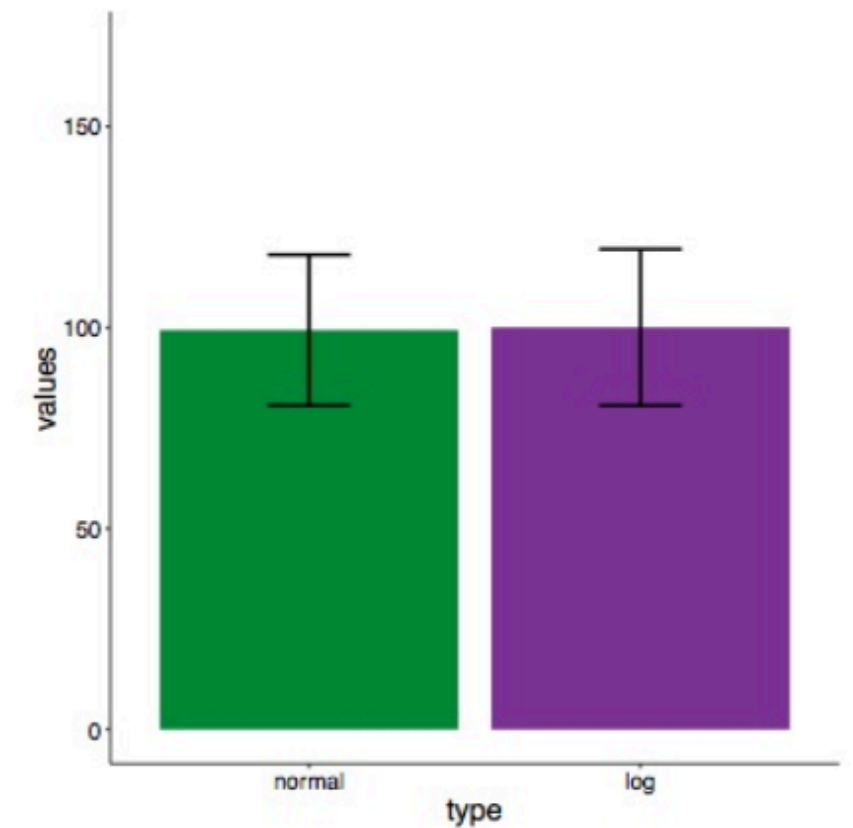
Clearly different...



also pretty clearly different....



the same!



GAME ANALITICS

Confidence interval

Relevant Questions

Understand standard deviation

understand the concept of Confidence interval

Be able to calculate both

Visualize confidence intervals